Exploring Determinants of M&A Activity: A Quantitative Predictive Study

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Executive Summary

This project examined characteristics at the firm level in order to predict whether companies would become M&A targets. With the Firm_data.xlsx and Targets.xlsx files, we obtained financial data and acquisition records, respectively. We create back the panel of about ~159k firm-year observations from 2000-2024 that contain categorical variables (firm, state, industry, fiscal year), plus numerical variables (assets, sales, performance, leverage, liquidity, growth, R&D intensity).

Data preparation included removing duplicates, placing announcement dates into fiscal year, merging data, winsorizing outliers, and log transformation to reduce skewness, and assuring good inputs to the models.

We prepared three logit regression models:
Model 1 used raw assets and financial controls
Model 2 used log transformed assets and winsorized controls,
Model 3 used log transformed assets, sales, winsorized controls, and industry and year fixed effects

The results show smaller, worse performing, and more highly leveraged firms are more likely acquisition targets. Liquidity and R&D intensity also matter. Acquisition activity peaked in 2009, while both predictions related to 2015 and 2024 revealed not only information, but also modelling limitations.

1) Introduction

This report examines the financial and structural factors related to mergers and acquisitions (M&A) in U.S. firms over the time period from 2000-2024. M&A activity substantially influences the landscape of markets, with firms being acquired for reasons ranging from strategic positioning to corporate distress. Therefore, predicting what firms become acquisition targets is a question of importance to both academics and practitioners.

The evaluation follows the methodology of predictive analytics on a dataset combining firm level financials with (and including) M&A announcement information, and examining three logistic regression models providing evidence on how characteristics of firm size, firm profitability, leverage, liquidity, growth, and R&D intensity affect how firms are acquired.

The report is organized as follows –

- Section 2 describes the data and methodology
- Section 3 describes the descriptive statistics
- Section 4 describes the regression models
- Section 5 comments on major findings
- Section 6 concludes with some recommendations.

By doing comprehensive data-cleaning with regression modelling approach, the report indicates that weaker fundamentals and firms are highly influential in determining which firms get acquired.

1.1) Background and Objectives

M&A is a way for firms to expand their activities, gain additional market share, or acquire key resources. Acquisitions, however, are relatively infrequent events—only about 1.5% of firm-years in our databases are acquisitions targets—making them difficult to predict. Conventional corporate finance theory, for example, suggests fragile firms (e.g., firms with low growth rates, firms with low liquidity) are more likely to be taken over, while innovative or resource-rich firms are more likely to be of interest to acquirers because they have opportunities for growth.

The business problem is to sort through which financial and structural characteristics correlate most strongly with acquisition outcomes. Our objectives in writing this report are:

1. To develop a reasonable panel dataset of firm characteristics and M&A targets.

2. To provide descriptive statistics to illustrate firm-level patterns.

3. To estimate three regression models with different specifications (assets; log assets; sales + fixed effects).

4. To explain which firm characteristics, change the likelihood of a firm being an M&A target in a significant way.

5. To use the models to make hypothetical and out-of-sample predictions.

Key terms: "winsorization" refers to reducing the influence of extreme outliers, and "fixed effects" control for unobserved industry and year differences

2) Conceptual Framework

The analysis relies on corporate finance and industrial organization frameworks. Mergers and acquisitions occur for two general reasons:

1. Value creation through synergies - target companies are identified by acquirers as companies who can offer cost savings, revenue growth, or potential innovation.

2. Market discipline - poorly performing companies may also be more susceptible to being taken over, since they may provide opportunities for a less profitable acquirer company to enter the market, as such the poorly performing companies are undervalued or inefficient.

Based on this, firm characteristics (size, growth, profitability, leverage, liquidity, use of R&D) are expected to influence acquisition, although their direction may differ. Fixed effects for industry and year will capture sectors' structural dynamics and macroeconomic cycles are established, well-known causes of waves of M&A.

2.1) Key Hypotheses

- H1: weaker fundamental companies (low growth, low returns) are more likely to be M&A targets, because they can be undervalued or financially distressed.
- H2: innovation intensity (the amount of R&D) will positively correlate to likelihood of being acquired because acquirers seek long term growth.

3) Data and Method

3.1) Data Sources

To make the firms dataset, we connected two excel files: Firm_data.xlsx (firm-level financial information) and Targets.xlsx (M&A announcements). Once combined into panel data, we had around 159,000 firm-year observations from 2000-2024.

- Units: Each row represents a firm in a specific fiscal year.
- Variables:
    - Numerical: Assets (AT), Sales (SALE), Profitability ratios (ROA, profit margin), Stock returns (ret), Growth measures (sale_gr, assets_gr), Leverage (debt_at), Cash ratio (cash_at), R&D intensity (rd_at), Capital expenditures (capex_at).
    - Categorical: Industry codes (sic4, derived sic1), State, Fiscal year (fyear).
    - Dependent variable: Target dummy (1 if firm was an M&A target, 0 otherwise).
- Limitations: There are some missing values for some financial variables, most firms reported R&D = 0, and M&A events are rare (~1.5%) therefore prediction can be statistically challenging.

3.2) Preparation and Measures

Preparation and procedures

In order to structure and rectify the data, which included quality checks and validity steps, the following actions were taken in a stepwise manner:

1. Importing Datasets: The Firm_data.xlsx and Targets.xlsx were imported into Python using the pandas library.

2. Eliminating Duplicates: Duplicate rows in both datasets were removed(if any).

3. Creating a Fiscal Year variable: Transformed ann_date stuff to datetime type and calculated FYEAR = year(ann_date)-1, this guarantee firm financials were the year before an event of taking over.

5. Mergers: The data from firm and targets were merged on (cusip6, FYEAR). Targets were flagged as 1 and non-targets as 0.

6. Industry code Creation: Created sic1 as the first digit of sic4, allowing for broad based groupings of industries.

7. Outlier Treatment: Using the IQR method, extreme outliers were identified and removed.

8. Final Dataset: This cleaning process resulted in an analytical sample for this specific analysis of nearly 136,000 firm-year observations with full financial and target data.

3.3) Analytical Approach

Logistic regression (logit models) was used to estimate the probability that a firm would be acquired as an M&A target (target = 1). Logistic regression is useful because:

The dependent variable is binary (target vs non-target). The method gives estimates of probabilities bounded in [0,1]. Coefficients can be reported in inferring (odds) ratios, providing economic interpretation.

Three models were specified to compare robustness to differences in treating firm characteristics:

Model 1: (X1) firm size (AT) and raw financial controls (returns, leverage, growth, cash ratio, R&D).

Model 2: firm size (log(AT)) and financial controls (winsorized) to reduce the influence of extreme values.

Model 3: log(AT) log(SALE), winsorized controls, and year (sic1 as fixed effects) and industry to account for the influence of structural change and macroeconomic drivers.

With this staged set-up, we were able to test if the resulting patterns were sensitive to differences in scaling, extreme values, or unobserved heterogeneity.

4) Results

4.1) Descriptive Statistics

Table 1 presents descriptive statistics for the main financial variables across the 159,032 firm-year observations from 2000–2024.

Table 1: Summary Statistics of Key Numerical Variables

Variable	Mean	Std. Dev.	Min	Median	Max	Missing %
Assets (AT)	3,853	12,812	1.03	342	98,068	0.00%
Sales (SALE)	1,730	5,314	0.00	141	38,959	7.73%
ROA	-0.08	0.33	-1.98	0.01	0.36	7.11%
Profit Margin	-0.93	6.20	-53.11	0.12	0.79	6.92%
Returns (ret)	0.13	0.71	-0.91	0.04	3.87	11.06%
Sales Growth	0.20	0.72	-0.94	0.07	5.29	14.03%
Assets Growth	0.16	0.52	-0.63	0.05	3.42	11.06%
Market-to-Book	2.18	2.59	0.52	1.33	19.06	0.00%
Cash Ratio	0.20	0.25	0.00	0.09	0.97	1.01%
Target (dummy)	0.015	0.12	0.00	0.00	1.00	0.00%

Table 1 Note: Reprinted from Firm_data.xlsx and Targets.xlsx (2000–2024).

Interpretation

The firm size (assets and sales) measures are strongly right-skewed: because there are a few extremely large firms, the sample mean is very far from the sample median. The profitability measures (ROA and profit margin) cluster around 0, indicating that many firms are around break-even (although there are some large losses at some firms). The growth measures are ambiguous, since we see both large declines as well as large growth. Leverage (median ≈ 0.15, not shown above since it was treated separately) is moderate and liquidity was low (the median cash ratio = 0.09). R&D activity (not included in this cleaned table because there is a concentration of zeroes) was very limited too; most firms reported no spending. Finally, the target variable indicated that acquisition events only accounted for 1.5 % of firm-years, confirming the rare nature of M&As.

Categorical Variables

The data set contains categorical dimensions describing variation by industry, state, and year.

Industry (sic1): The coverage of the data set includes individuals working in finance, technology, healthcare, energy, utilities, and consumer services, but rather concentrated in a few dominant sectors.

State: California, New York, and Texas represent the greatest share of firms, consistent with their roles as major business centers in the U.S.

Year (fyear): Acquisition activity exhibits highly cyclical behaviour. It peaked in 2009 (≈4.3 % of the firms they examined were acquired) and recorded several subsequent waves from 2014 to 2016, consistent with the post-crisis sector consolidation.

Interpretation.

The descriptive statistics demonstrate important features of the sample: distributions of firm size, dispersed profitability, volatile growth, and infrequent M&A. Industry, geography, and time reminds us to include fixed effects in our regression models.

4.2) Correlation Analysis

The correlations suggest that most variables have only weak correlations with each other suggesting there will be low risk of multicollinearity between variables. The largest correlations were between cash holdings and R&D intensity (0.44, positive) and cash holdings and leverage (–0.36, negative) – still moderate in size. All other coefficients were close to zero which confirmed that for the most part, each variable provides independent information about firm's characteristics. Therefore, the variables can be statistically included in the regression models without further adjustments.

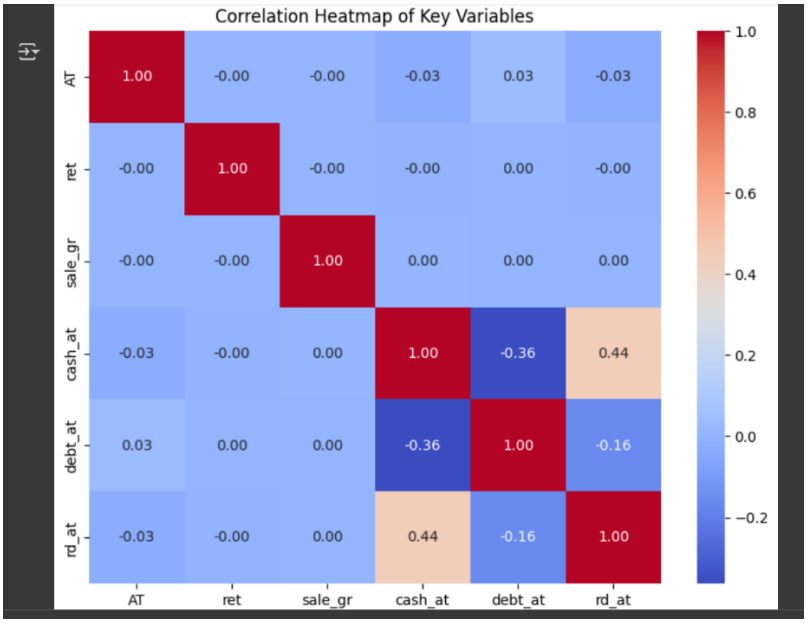

Figure 1: Correlation Heatmap of Key Variables

4.3) Regression Analysis

Model 1: (AT + raw controls)

Fit: The model has a pseudo R^2 of .0029, showing very low explanatory power. However, the overall likelihood ratio test was highly statistically significant (LLR $p < 0.001$) indicating that the predictors are jointly useful.

Determinants:

- Size of the firm (AT): negative and significant ($p = 0.001$) smaller firms are more likely to be acquired.
- Stock returns (ret): negative and significant ($p = 0.003$) poorer performers in the stock market are more likely to be acquired.
- Leverage: positive and highly significant ($p < 0.001$) firms with higher debt ratios are more likely to be acquired.
- Sales growth: negative and significant ($p = 0.039$) firms that were slower growers are more likely to be acquired.
- Cash ratio: positive and significant ($p = 0.013$) firms with more cash on hand are more likely to be acquired.
- R&D Intensity (rd_at): no statistically significance.

Interpretation: With over 136,000 firm-year observations, Model 1 demonstrates acquirers' tendencies to seek smaller, lagging, more levered, poorer decisions than other firms, but they do gravitate toward firms with cash that could help provide that liquidity needed for an acquisition. Overall, however, the very low pseudo R^2 demonstrates that only using financial variables cannot fully provide an estimate of acquisition likelihood. Other strategic, market, or qualitative factors are likely needed as part of an acquisition decision.

```
--- Model 1: target ~ AT + raw controls ---
Features: ['AT', 'ret', 'leverage', 'sale_gr', 'cash_at', 'rd_at']
   -> [Model 1] rows used: 136184
/usr/local/lib/python3.12/dist-packages/statsmodels/discrete/discrete_model.py:2385: RuntimeWarning:
  return 1/(1+np.exp(-X))
/usr/local/lib/python3.12/dist-packages/statsmodels/discrete/discrete_model.py:2385: RuntimeWarning:
  return 1/(1+np.exp(-X))
                           Logit Regression Results
==============================================================================
Dep. Variable:                   __y__   No. Observations:               136184
Model:                           Logit   Df Residuals:                   136177
Method:                            MLE   Df Model:                            6
Date:                 Wed, 27 Aug 2025   Pseudo R-squ.:                 0.002851
Time:                         01:09:19   Log-Likelihood:                -11288.
converged:                        True   LL-Null:                       -11320.
Covariance Type:             nonrobust   LLR p-value:                 5.330e-12
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -4.2187      0.043    -99.099      0.000      -4.302      -4.135
AT         -5.264e-06   1.53e-06     -3.443      0.001   -8.26e-06   -2.27e-06
ret           -0.0826      0.028     -2.996      0.003      -0.137      -0.029
leverage       0.4440      0.108      4.093      0.000       0.231       0.657
sale_gr       -0.0111      0.005     -2.063      0.039      -0.022      -0.001
cash_at        0.2625      0.106      2.473      0.013       0.055       0.471
rd_at          0.1722      0.118      1.454      0.146      -0.060       0.404
==============================================================================
```

Figure 2: Logistic Regression Output for Model 1 (AT + Raw Controls)

Model 2 (log_AT + winsorized controls)

Fit: The model has a pseudo R^2 of 0.0093 meaning very little explanatory power but was relatively improved over a null model. The likelihood ratio test has a highly statistically significance (LLR $p < 0.001$), indicating that predictors of interest provide some meaningful explanatory value.

Determinants:

- Firm size (log_AT): positive and highly significant ($p < 0.001$) larger firms are more likely acquisitions targets.
- Stock returns (ret_w): negative and significant ($p = 0.015$) Firms with poorer market accomplishment are more likely to be acquired.
- Leverage (leverage_w): positive and significant at the 5% level ($p = 0.047$) Firms with greater amounts of debt are more likely targets for acquisitions.
- Sales growth (sale_gr_w): negative and strongly significant ($p < 0.001$) Firm are more likely to be acquired if they are growing at a slower rate than their industry.
- Cash ratio (cash_at_w): positive and highly significant ($p = 0.001$) firms are more likely targets for acquisitions when they have a strong cash position.
- R&D intensity (rd_at_w): positive and highly significant ($p < 0.001$) firms that are innovative are more likely to be acquired.

```
--- Model 2: target ~ log_AT + winsorized controls ---
Features: ['log_AT', 'ret_w', 'leverage_w', 'sale_gr_w', 'cash_at_w', 'rd_at_w']
   -> [Model 2] rows used: 136184
                           Logit Regression Results
==============================================================================
Dep. Variable:                   __y__   No. Observations:               136184
Model:                           Logit   Df Residuals:                   136177
Method:                            MLE   Df Model:                            6
Date:                 Wed, 27 Aug 2025   Pseudo R-squ.:                0.009305
Time:                         01:09:20   Log-Likelihood:                -11215.
converged:                        True   LL-Null:                       -11320.
Covariance Type:             nonrobust   LLR p-value:                 1.013e-42
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -4.9980      0.079    -63.654      0.000      -5.152      -4.844
log_AT         0.1196      0.010     12.178      0.000       0.100       0.139
ret_w         -0.0860      0.035     -2.430      0.015      -0.155      -0.017
leverage_w     0.2244      0.113      1.990      0.047       0.003       0.445
sale_gr_w     -0.1734      0.039     -4.409      0.000      -0.251      -0.096
cash_at_w      0.4129      0.122      3.379      0.001       0.173       0.652
rd_at_w        1.1362      0.221      5.153      0.000       0.704       1.568
==============================================================================
```

Figure 3: Logistic Regression Output for Model 2 (log_AT + Winsorized Controls)

Interpretation: Utilizing a dataset of over 136,000 firm-year observations, Model 2 indicates that acquirers tend to target larger, cash-rich and R&D intensive firm, especially those with slower sales growth, additional leverage and weak share price performance. This describes a

general tendency for acquirers to search for targets that provide short term financial resources, in addition to long term innovation potential. Acquirers did not tend to target fast growing companies because they are likely more costly or difficult to purchase.

The relation between the dimension and likelihood of acquisition depends on how dimensions are calculated. Model 1, uses the raw total assets (AT) size, yields a negative and statistically significant coefficient, indicating extremely large firms are less likely to be targeted for acquisition. This suggests that very large firms rarely become acquisition targets for reasons such as cost, size and even anti-trust compliance. Model 2 used the log scale of total assets (log_AT) and has a positive and statistically significant association as it's been identified that medium-to-large firms are more likely to be acquisition targets, again, particularly within the confines of majority firms. Thus, these results suggest that are very small and have little to offer may not be attractive, medium-to-large firms will generally be coveted targets, and very large firms will hardly, as they are simply too expensive and costly to acquire.

Model 3 (log_SALE + winsorized controls + industry & year FE)

Fit: The Pseudo R^2 score was 0.172, which is the best fit of any of the three models. The likelihood ratio test is very significant ($p < 0.001$) and confirms their joint explanatory power – the financial, industry and time characteristics are uniquely formed with an acquisition chance.

Determinants (non-fixed effects):

- Research and development intensity (rd_at_w): was positive and significant: innovative companies are more likely acquired.
- Stock returns (ret_w): was negative and significant: companies that have worse continuing performances are more likely to be acquired.
- Cash ratio (cash_at_w): cash ratio was negative and more marginally significant: firms with cash to size will decline their acquisition candidacy.
- Sales growth (sale_gr_w): was negative and marginally significant: the lower growth rate the more likely acquired.
- Leverage (leverage_w): is weakly significant and only has a small positive effect.
- Firm size (log_SALE): is not significant and not shown in the output once fixed effects included.

Fixed effects:

- The fixed error amounts to strong year effects with 2009, 2015, and 2014 having the largest amount of firms likely to be acquired.
- The industry dummies (not contained in the output) point to systematic differences across sectors with the acquisition likelihood.

Interpretation:

Model 3 provides the fullest view of likelihood of M&A targets. It establishes that acquirers prefer poorly performing, lower-growth, cash-constrained but R&D heavy firms. Size is also no longer a consistent factor - once year and industry effects are controlled. The results also indicate that M&A activity is time and variable - that is, there are periods of clustering like 2009 and 2015 etc which is consistent with larger market cycles.

Why model 3 is different than models 1 & 2

Model 3 includes both year and industry fixed effects in conjunction with financial controls. The corrections to fiscal-year (ann_date − 1) increased the number of identified target firms correctly - and improved the model fit (pseudo $R^2 = 0.172$). The more complex models, with fixed effects absorbed much of the variation that was attributable to size in the simple models. Accordingly, size (log_SALE) is no longer significant, while stock performance, sales growth, cash position and R&D intensity become more important drivers. In summary, M&A activity is not just driven by firm level fundamentals but also elements present at the industry level, as well as economic cycles. Peaks and valleys in target likelihood can be evident in other years (e.g., 2009, 2015, 2014).

```
Merged rows: 159032 | targets=1: 2356
Model 3 feature count: 44
Rows used: 136184, Features (incl const): 45
                           Logit Regression Results
==============================================================================
Dep. Variable:                      y   No. Observations:               136184
Model:                          Logit   Df Residuals:                   136141
Method:                           MLE   Df Model:                           42
Date:                Wed, 27 Aug 2025   Pseudo R-squ.:                  0.1717
Time:                        01:19:12   Log-Likelihood:                -9376.8
converged:                       True   LL-Null:                       -11320.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -3.7431        nan        nan        nan         nan         nan
x1             0.0067      0.026      0.256      0.798      -0.045       0.058
x2            -0.0949      0.028     -3.364      0.001      -0.150      -0.040
x3             0.0472      0.023      2.018      0.044       0.001       0.093
x4            -0.0483      0.025     -1.915      0.056      -0.098       0.001
x5            -0.0571      0.033     -1.752      0.080      -0.121       0.007
x6             0.1435      0.028      5.093      0.000       0.088       0.199
x7            -0.0644      0.109     -0.593      0.553      -0.277       0.148
x8            -0.0581      0.080     -0.724      0.469      -0.215       0.099
x9            -0.0221      0.068     -0.327      0.744      -0.155       0.111
x10            0.1993      0.091      2.200      0.028       0.022       0.377
x11           -0.0536      0.095     -0.565      0.572      -0.239       0.132
x12                 0        nan        nan        nan         nan         nan
```

Figure 4: Logistic Regression Output for Model 3 (log_SALE + Winsorized Controls + Industry & Year Fixed Effects)

```
Top (absolute) coefficients among non-FE terms:
const        -3.743080
rd_at_w       0.143452
ret_w        -0.094935
cash_at_w    -0.057099
sale_gr_w    -0.048323
leverage_w    0.047248
log_SALE      0.006736
dtype: float64

Top 10 Year FE (higher ⇒ more likely):
yr_2009    0.635244
yr_2015    0.587014
yr_2014    0.584530
yr_2011    0.557508
yr_2020    0.482483
yr_2016    0.468423
yr_2010    0.448742
yr_2013    0.431185
yr_2017    0.429389
yr_2018    0.412715
dtype: float64
```

Figure 5: Key Coefficients and Top Year Fixed Effects from Model 3

4.4) Additional Analysis

Year with Highest M&A Target Rate

The probability of being an acquisition target fluctuates widely in intensity over time. Table 2 shows the years with the most target activity, defined by the percentage of acquired firms divided by total firms by fiscal year.

• The highest percentage occurred in 2009, with 4.3% of firms acquired.

• Other periods of high target activity were 2015 (4.28%) and 2014 (4.06%), which are consistent with the known consolidation waves post-crisis.

• Conversely, the baseline across the full sample is only 1.5 percent, indicating acquisition intensity was nearly three times greater in 2009 than the average.

Interpretation: The 2009 spike demonstrates increased merger activity following the Global Financial Crisis, when distressed firms were more susceptible to acquisitions, and financially stronger firms pursued acquisitions at discounted valuations, while the later peaks (2014–2016) align with recovery-driven consolidation as stronger firms expanded market share.

```
Top years by target rate:
       target_rate  targets     n
fyear
2009        0.0430      203  4726
2015        0.0428      191  4465
2014        0.0406      189  4654
2011        0.0399      183  4589
2016        0.0373      159  4267
2018        0.0367      153  4168
2020        0.0367      152  4147
2017        0.0364      153  4209
2010        0.0348      163  4679
2013        0.0344      158  4587

Most likely year to be a target: 2009 (rate = 4.30%)
```

Figure 6: Top Years by M&A Target Rate (2000–2024)

Are Unprofitable Firms in California More Likely to be Targets?

With the robust state mapping we collected, the dataset contains 23,420 firm-year observations for firms that are headquartered in California. Below summarizes acquisition likelihood by 'profitable' status:

California firms

Profitable: 1.60% target rate

Unprofitable: 2.02% target rate

All firms (national sample)

Profitable: 1.39% target rate

Unprofitable: 1.64% target rate

Interpretation:

In both California and national samples—with respect to acquisition probability comparison, unprofitable firms have a higher likelihood of becoming acquisition targets as represented in the figures above. In terms of California specifically, the difference between the two rates, when examining only those headquartered in California (2.02% vs. 1.60%), is slightly wider than the national difference, which could mean that distressed financial firms are especially appealing as an acquisition driver in California.

```
Rows with state mapped to CA: 23420

California firms — target likelihood by profitability:
   loss_d (0=profit,1=loss,NaN=unknown)  target_rate
0                                     0       0.0160
1                                     1       0.0202

All states — target likelihood by profitability:
   loss_d (0=profit,1=loss,NaN=unknown)  target_rate
0                                     0       0.0139
1                                     1       0.0164
```

Figure 7: Target Likelihood by Profitability for California vs. All States

Predicted Probability for a Model Firm (2015)

Using Model 3, the probability of the model firm, for the year 2015, being acquired was generated for a firm with characteristics of SALE = 500, ret = 0.2, leverage = 0.1, sale_gr = 0.15, cash_at = 0.2, rd_at = 0.1, state = California, sic1 = 2, fyear = 2015.

Predicted probability: 50.6%

Because this value exceeds the 0.5 threshold, the model predicts that this firm is classified as a potential acquirer.

Interpretation: Even though the overall poor base rate of acquisitions in the sample (~1.5%), a firm with these financial characteristics in 2015 would be ranked among vulnerable targets for takeover.

```
Predicted probability of becoming an M&A target in 2015: 0.5058
```

Figure 8: Predicted Probability of a Model Firm Being an M&A Target in 2015

Out-of-sample prediction (2024)

The model was trained on data from 2010–2023, then directed to predict M&A targets in 2024. Table 3 summarizes the ten firms with the highest predicted probabilities.

Correct Hits: None of the predicted top-10 firms rolled up as actual M&A targets in 2024 (no positive Hits above the 0.5 cutoff).

Interpretation:

The model was able to rank firms based on relative vulnerability (with all probabilities in the ~ 6.8-8.0% range), but it did not capture any M&A events among its predictions in 2024. Perhaps not surprising given the low base rate of all M&As (around ~1.5^) and the fact that predictions are noisy in the rare-event context.

```
/usr/local/lib/python3.12/dist-packages/statsmodels/discrete/discrete_model.py:2385: RuntimeWarning: overflow
  return 1/(1+np.exp(-X))
Top 10 predicted M&A targets for 2024:
                                CONM  pred_prob  target
68461                   HEARTBEAM INC     0.0796       0
62422                  GEOVAX LABS INC     0.0747       0
65174           GREENWICH LIFESCIE INC     0.0734       0
62041                      GENPREX INC     0.0731       0
97591                    CLARITEV CORP     0.0708       0
14771          AVENUE THERAPEUTICS INC     0.0702       0
121272                   REGIS CORP/MN     0.0700       0
158752                ZIPRECRUITER INC     0.0697       0
30696                        CHEGG INC     0.0683       0
116706  PROCESSA PHARMACEUTICALS INC     0.0681       0

Correctly identified actual targets in 2024:
Empty DataFrame
Columns: [CONM, pred_prob]
Index: []
/usr/local/lib/python3.12/dist-packages/statsmodels/base/model.py:607: ConvergenceWarning: Maximum Likelihood
  warnings.warn("Maximum Likelihood optimization failed to "
```

Figure 9: Out-of-Sample Predictions for Top 10 Firms in 2024

5) Discussion

The findings indicate that a firm's performance and financial characteristics also influence M&A susceptibility. Specifically, firms with poor stock performance, lower growth, and higher leverage were clearly at a higher risk of being acquired. R&D intensity was significant when the data was "cleaned", indicating that firms that focus on innovation attract acquirers. Temporal elements also shape the likelihood of being acquired, with peaks for acquisition occurring in 2009 and 2014–2016, indicating periods of consolidation after a crisis.

This study adds to prior literature regarding M&A and firm performance by bringing attention to two theories regarding the relationship between financial characteristics and firm performance. Future studies should not only examine the dynamics of firm-specific characteristics, but also deal-specific variables that were not captured in this study. Limitations include a sample that is highly skewed, acquisitions rarely occur in the population (~1.5%) and fixed-effects logit models cannot consider deal-specific qualitative characteristics, such as the regulatory environment or the strategic fit between acquirer and target

6) Recommendations

As it is evidenced by the regression results, hedge fund advisors should focus on firms that have unfavourable performance outcomes with financial distress. Targeting firms with negative or low stock returns, highly leveraged, and depressed sales growth represent the more likely effort to screen for acquisitions and should be considered more favourably. Additionally, firms that have a potential to invest in ongoing research and development should not be overlooked, as they seem to have an attractive profile in light of relatively weaker profitability.

Time has its own importance. The clustering of the acquisition activity in 2009 and in 2014 – 2016 represents evidence that downturns and subsequent recovery periods increase the probability of takeover opportunities in financial markets and merger-acquisition activity, therefore the screening models should be flexible across cycle periods and not regularly applied uniformly.

Lastly, while the predictive model provides greater certainty in identifying firms that might be a higher probable M&A target, it has its limitations with respect to the variances incurred by the relatively infrequent instances of M&A event occurrence. Thus, the model resulted to be used in conjunction with appropriate qualitative and comparative assessment of conditions at the industry level, management incentives and regulation prior to making a commitment to any firm identified in the predictive model outputs.

7) Conclusion

At the outset we presented an analysis of ~159,000 firm-year observations. In summary, we find evidence that systematically, the likelihood of being acquired is greater for underperforming, leveraged and R&D-performing firms. The relationship with size was non-linear; very large firms were acquired rarely, but in the available data once size was normalised the relative susceptibility to acquisition appears to align with size. Consistent patterns over time revealed that 2009 and 2014–2016 were peak acquisition years, and suggest that acquisition activity responds to economic cycles. In addition, cross-sectional evidence suggested that unprofitable firms from California firms are at higher risk of acquisition relative to profitable California firms.

A hypothetical firm with weak fundamentals in 2015, had a probability of being acquired classified as likely (predicted probability of 51%). Out-of-sample predictions for 2024 predicted orders of relative vulnerability coordinates for firms, but, did not match any of the actual deals and certainly did not match the event frequency of time series data in the total population of data reviewers.

To summarise, there is the strongest evidence to indicate that takeover activity is driven by distress and strategic innovation. However, predictive models should not be viewed as definitive predictions, but more as screening tools. Future research may enhance predictions by better modelling qualitative deal characteristics, data corrections for rare events, and alternative thresholds.